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Abstract

This study analyzes the *Superstore* dataset from the perspective of a manager focused on optimizing corporate profitability: through the integration of Statistical Learning techniques, the study explores the dynamics between logistics, discounts, and profits to identify the drivers that generate economic losses. Unsupervised learning models are applied for customer segmentation, and supervised algorithms are used for transactional risk classification. The analysis aims to transform complex data into a solid scientific basis for revising pricing and distribution strategies.

The analysis reveals that the discount (`Discount`) is the dominant driver of profitability, explaining on average 84% of the variance in the tested models. Additional critical determinants include the *Furniture* product category, sales volumes, and geographic location in the *Central* region.

In terms of model performance, *XGBoost* demonstrated the highest predictive capability; however, *LASSO* regularization highlighted how a parsimonious structure of only four variables is sufficient to explain the phenomenon with high accuracy.

In conclusion, the project demonstrates how adopting an analytical approach makes it possible to identify inefficiencies that are invisible to a simple superficial reading of the data, providing a solid scientific basis for revising corporate pricing and distribution strategies.

Chapter 1

Introduction

1.1 Context and Objectives

In an era dominated by data, the ability to transform simple rows of transactions into actionable insights is what enables a manager to evaluate the effectiveness of pricing policies, optimize logistics, and maximize marginal profits on a national scale. In the retail sector, the analysis of transactional data is a fundamental pillar for improving operating margins.
The strategic question this research aims to answer is:

> ***Which variables most influence the risk of generating an economic loss on a single transaction?***

1.2 Methodology of Analysis

The analytical process follows a sequential structure in three phases:

1. **Exploratory Data Analysis (EDA):** Aimed at data cleaning, graphical exploration of the data and variables, and identification of initial correlations between variables.

2. **Unsupervised Learning:** Use of PCA (Principal Component Analysis) to reduce the dimensionality of the analysis and clustering to group transactions into homogeneous profiles.

3. **Supervised Analysis:** Implementation of predictive models to classify transactions using the binary target variable `loss_flag`. The multi-model approach ensures the robustness of the conclusions, making it possible to distinguish between simple correlations and operationally actionable business drivers.

1.3 Dataset Description

The dataset under analysis represents a strategic information asset for understanding the commercial dynamics of a large American retailer. By analyzing the structure of the `Superstore` file, the following dimensional characteristics emerge:

- **Number of rows:** 9,994 (individual transactions);
- **Number of columns:** 13 (main extracted variables);
- **Geographic coverage:** 4 Regions (Central, East, South, West) and 48 U.S. states.

Variable Taxonomy

To proceed with a rigorous analysis, it is useful to classify the 13 variables based on their functional nature:

1. **Customer and Geographic Profiling:** fundamental for marketing and market segmentation.
   - `Segment`: Customer segment (Consumer, Corporate, Home Office);
   - `Country`, `City`, `State`, `Postal Code`, `Region`: Geographic hierarchy to identify more or less profitable markets;
2. **Logistics Variables:** this information allows analysis of the shipping methods used in the supply chain.
   - `Ship Mode`: Shipping method (e.g. Second Class, Standard Class, First Class, Same Day);
3. **Product Characteristics:** enable analysis of the sales mix and performance by category.
   - `Category`: Macro-category (Furniture, Office Supplies, Technology);
   - `Sub-Category`: Specific sub-category (e.g. Tables, Phones, Art);
4. **Economic Metrics (Quantitative):** the core of financial analysis, useful for calculating KPIs or operating margin.
   - `Sales`: Gross sales value;
   - `Quantity`: Number of units sold;
   - `Discount`: Discount percentage applied;
   - `Profit`: Net profit derived from the transaction;

Chapter 2

Exploratory Data Analysis (EDA)

2.1 Data Cleaning and Preparation

The dataset was subjected to a *Data Cleaning* process to simplify the analysis and optimize the accuracy of statistical models:

- **Dimensionality reduction:** removal of the variables `country` (constant for the United States) and `postal_code` to avoid geographic redundancies.
- **Granularity simplification:** the variable `product_name` was discarded in favor of `category` and `sub_category` to avoid excessive data fragmentation.
- **Typing:** conversion of textual variables (such as `ship_mode` and `segment`) into factors to facilitate computational handling.

2.2 Variable Analysis and Visualization

2.2.1 Correlation between Numerical Variables

The quantitative analysis (Figure 2.1) revealed the fundamental interdependencies among financial metrics:

- **Sales vs Profit (0.48):** there is a moderate positive correlation; increasing sales tends to increase profit, although not in a perfectly linear way.
- **Discount vs Profit (-0.22):** a negative correlation is observed, confirming that high discounts directly impact margins.
- **Quantity:** shows a weak correlation with profit, indicating that the volume of items sold does not guarantee higher earnings without an adequate unit margin.

The correlation matrix shows that there are no collinearities requiring a reduction in the number of numerical variables considered, allowing all of them to be retained for subsequent analyses.

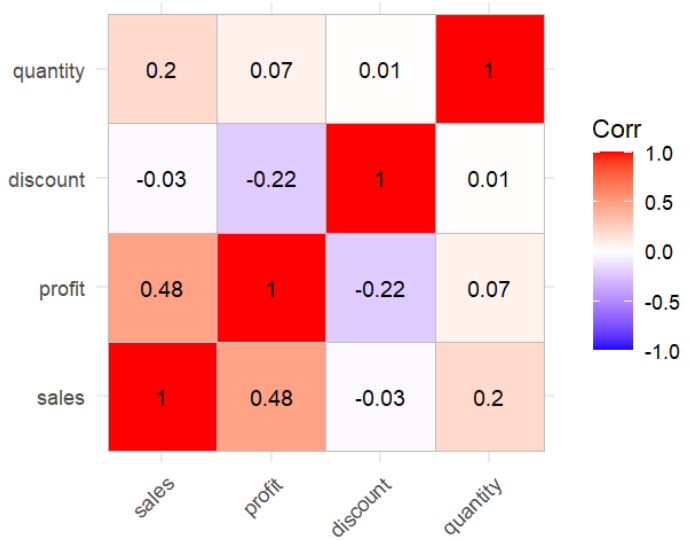

Figure 2.1: Correlation matrix for numeric variables

2.2.2 Loss flag visualization

We proceed with the analysis of the distribution of transactions in terms of profits and losses (Figure 2.2). To this end, each transaction is classified using the binary variable `loss_flag`; this labeling will be instrumental for training supervised models for the predictive classification of financial outcomes.

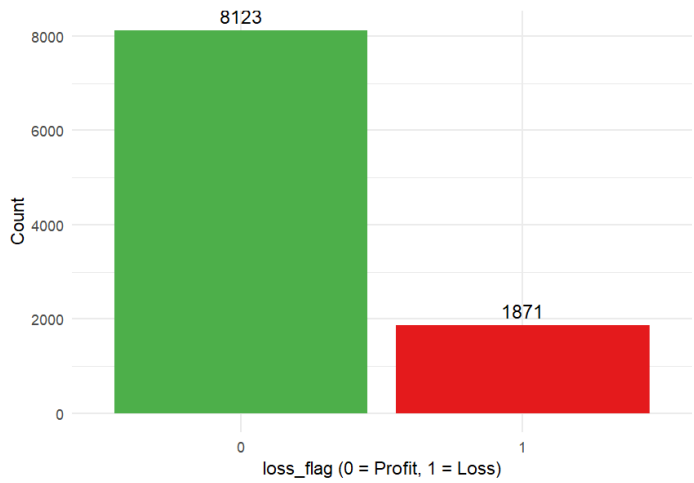

Figure 2.2: Profit and loss proportion of target variable

The number of loss-making transactions is lower than that of profitable ones, but still accounts for around 20% of the total. This result calls for a more in-depth analysis to investigate the reasons behind such a high proportion of losses.

2.3 Profitability Drivers and Critical Issues

Having completed the preliminary analysis on the dataset structure, variable nature, and their correlations, it emerged that the incidence of loss-making transactions represents a significant issue. Consequently, it is necessary to investigate which additional factors and exogenous dynamics influence profitability and, more broadly, the company's financial stability.

The "Discount" Effect

The analysis of the relationship between profit and applied discounts, illustrated in (Figure 2.3), makes it possible to identify a 'critical threshold' beyond which profitability becomes systematically negative. Empirical evidence shows a marked deterioration in margins for discounts above 20–30%, identifying a clear tipping point for the economic health of transactions.

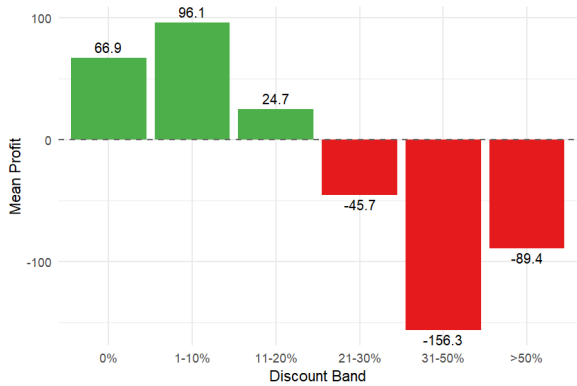

Figure 2.3: Discount buckets and mean profit

Performance by Product Category

The analysis continues by examining potential heterogeneity in results across different product categories (Figure 2.4). The objective is to isolate segments that generate systematic losses and evaluate the contribution of each category to overall profitability and business continuity.

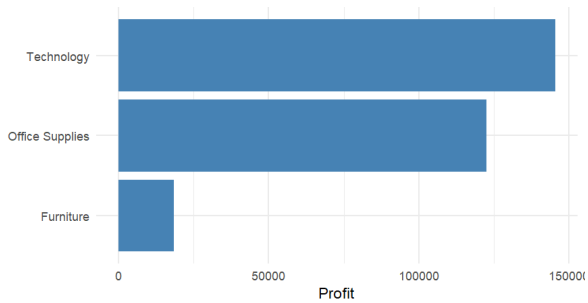

Figure 2.4: Profit per category

- **Furniture:** emerges as the most structurally problematic category, with low margins and frequent losses;
- **Technology:** despite lower volumes, generates significant profits thanks to the high unit value of some products;
- **Office Supplies:** represents the core business in terms of consistent transaction volume.

Geographic and Logistics Distribution

In line with what was introduced earlier, the dataset aggregates geographic units into four macro-regions. To optimize model dimensionality and simplify the analysis, `region` was selected as the main geographic reference variable. Below, the distribution of data across these areas is illustrated (Figure 2.5) to verify its consistency and structure.

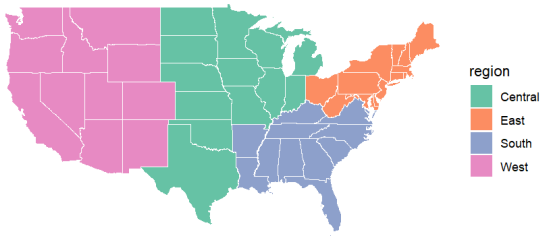

Figure 2.5: Region partition

- **West Region:** consistently emerges as the most profitable geographic area (Figure 2.6).
- **Central Region:** shows a specific anomaly with below-average profitability, regardless of product type, suggesting inefficient cost management or discount policies in this area (Figure 2.6).

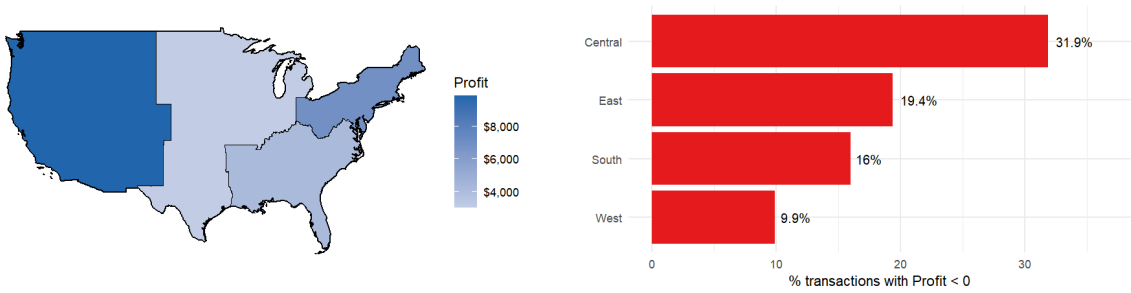

Figure 2.6: Region analysis. *Left*: Profit per region. *Right*: loss percentage of transaction by region.

Chapter 3

Unsupervised Analysis

After outlining the exploratory profile of the data, the analysis proceeds with the application of unsupervised learning techniques. The objective of this phase is not the prediction of a specific variable, but rather the exploration of the latent structure of the dataset to identify how variables move jointly and which operational profiles characterize the *Superstore* dataset.

3.1 Principal Component Analysis (PCA)

Through Principal Component Analysis (PCA), the original dataset is projected into a new space of uncorrelated principal components, summarizing the input variables (*Sales, Quantity, Discount, Profit*). This approach is crucial for isolating latent drivers underlying economic success or operational inefficiencies, providing a more streamlined and immediately interpretable informational structure.

3.1.1 Theoretical Foundations and Methodology

From a conceptual standpoint, PCA is an *unsupervised learning* method that allows summarizing a large set of correlated variables while minimizing information loss (variance). The first principal component is the linear combination of the original features that maximizes variance:

$$z_1 = a_{11}x_1 + a_{21}x_2 + \cdots + a_{p1}x_p \tag{3.1}$$

subject to the normalization constraint of the weights (loadings) $\sum_{j=1}^{p} a_{j1}^2 = 1$. Subsequent components (z_m) maximize the residual variance with the additional constraint of being **orthogonal** to the previous ones, ensuring the absence of multicollinearity among the new synthetic predictors.

3.1.2 Component Selection and Explained Variance

A crucial step concerns the choice of the *Proportion of Variance Explained* (PVE), evaluating the trade-off between model simplicity and information retention.

The effectiveness of dimensionality reduction in our case is confirmed by the **Scree Plot** (Figure 3.1):

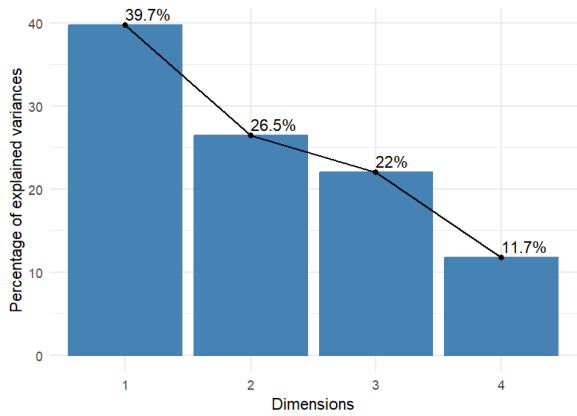

Figure 3.1: Variance Distribution by Dimension

- The first two components (PC1 and PC2) explain approximately 66% of the total variance (PC1: $\sim 40\%$, PC2: $\sim 26\%$).
- Including the third component (PC3), nearly 90% of the information is captured, indicating that transaction complexity can be summarized in an extremely compact structure.

Based on the analysis of loading scores (Figure 3.2), only the first two principal components were retained. Although the third component contributes to cumulative variance, it does not introduce new interpretative dimensions, appearing as a redundant linear combination of the first two variables. The 66% variance explained by PC1 and PC2 alone was therefore considered an optimal trade-off between parsimonious synthesis and fidelity to the original data.

Economic Interpretation of Components (Loadings)

Through the analysis of weights (*loadings*), it is possible to assign a business meaning to the identified components:

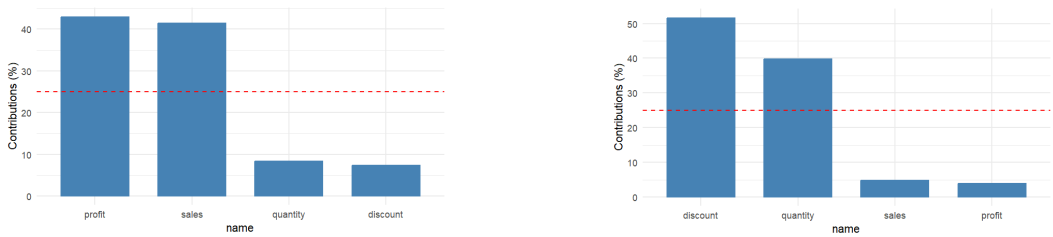

Figure 3.2: Contribution of variables to the first two dimensions. *Left*: First component. *Right*: Second component.

- **PC1 – The Performance Dimension:** dominated by *sales* and *profit*, which exhibit high and consistent loadings. This component summarizes the magnitude of commercial success: a positive movement along this axis reflects a joint increase in revenue and margin.

- **PC2 – The Volume Strategy Dimension:** dominated by `discount` and `quantity`. This component captures promotional dynamics. It emerges that discount has a very strong influence, often opposing the pure profitability captured by PC1.

PCA Plane Analysis (Biplot)

The projection of variables onto the plane defined by the first two principal components provides a geometric synthesis of the interdependencies identified during the exploratory analysis (EDA). The Biplot (Figure 3.3) allows visualization not only of the dataset structure but also of directions of maximum variability:

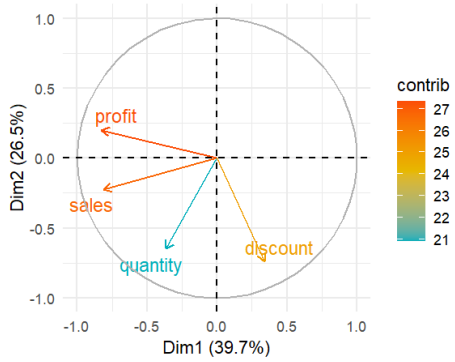

Figure 3.3: PCA plane analysis (Biplot)

- **Profit-Revenue Synergy:** The vectors *sales* and *profit* show strong angular proximity, confirming a direct positive correlation and the ability of PC1 to capture operational profitability.
- **Discount-Margin Trade-off:** The *discount* vector is positioned almost opposite to *profit*. This angular divergence graphically illustrates how aggressive discount policies systematically reduce margins.
- **Volume-Promotion Relationship:** The similar orientation of *quantity* and *discount* along the PC2 axis suggests that high volumes are typically driven by price incentives rather than organic value growth.

Managerial Implications

The analysis reveals that the Superstore business is polarized between a "revenue-driven growth" (PC1) and a "volume-through-discounts strategy" (PC2). Beyond providing a clear visualization of transactions in a low-dimensional space, this process serves a broader purpose:

1. **Noise reduction:** Only the dominant data structure is retained.
2. **Clustering input:** The coordinates of the first two components are used as input features for the segmentation algorithm, ensuring more stable and interpretable clusters.

3.2 Clustering

Leveraging the geometric space defined by the first two principal components, the analysis proceeds with transaction segmentation using clustering techniques. The objective is to uncover the latent structure of the dataset, transforming a multidimensional cloud of data into well-defined strategic categories.

This approach addresses two fundamental needs:

- **Pattern abstraction:** Moving beyond univariate analysis to observe how variables combine in determining transaction outcomes.
- **Profiling:** Identifying macro business types and behavioral patterns that would not be visible through traditional analysis.

3.2.1 K-Means Clustering

K-means aims to partition observations into K distinct and non-overlapping clusters. The algorithm minimizes within-cluster variance (Within-Cluster Sum of Squares, WCSS), defined as:

$$\sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

where C_k represents the k-th cluster and μ_k is its centroid.
To determine the optimal number of clusters K, the *Elbow Method* (Figure 3.4) was used, testing values from 1 to 10. The WCSS curve shows a clear "elbow" at $K = 4$.

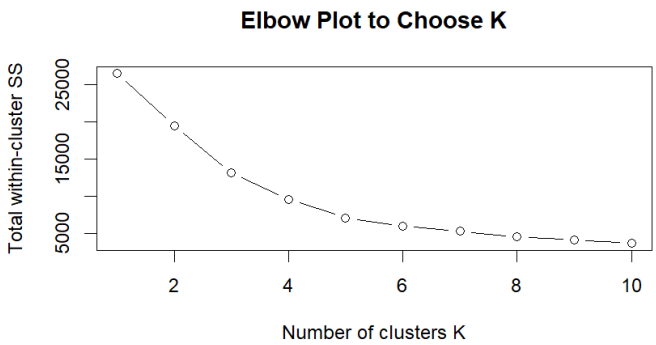

Figure 3.4: Elbow plot

Segmentation with K-Means

With the optimal configuration ($K = 4$), the dataset was segmented using the K-means algorithm. A quantitative analysis based on averages (numerical variables) and frequencies (categorical variables) was then conducted, supported by visual analysis (Figure 3.5)(Figure 3.6) to highlight morphological patterns and structural divergences between groups.
Each cluster was assigned a descriptive business-oriented label to facilitate interpretation and application.

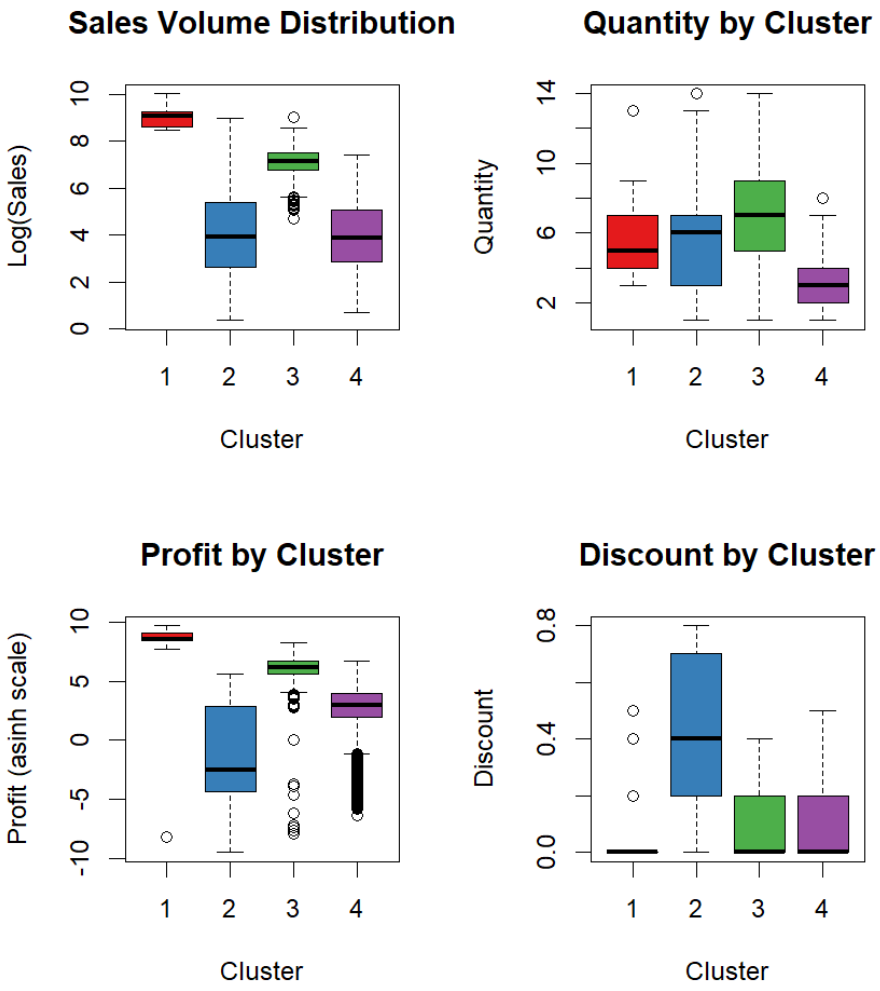

Figure 3.5: Cluster characterization and contribution of numerical variables.

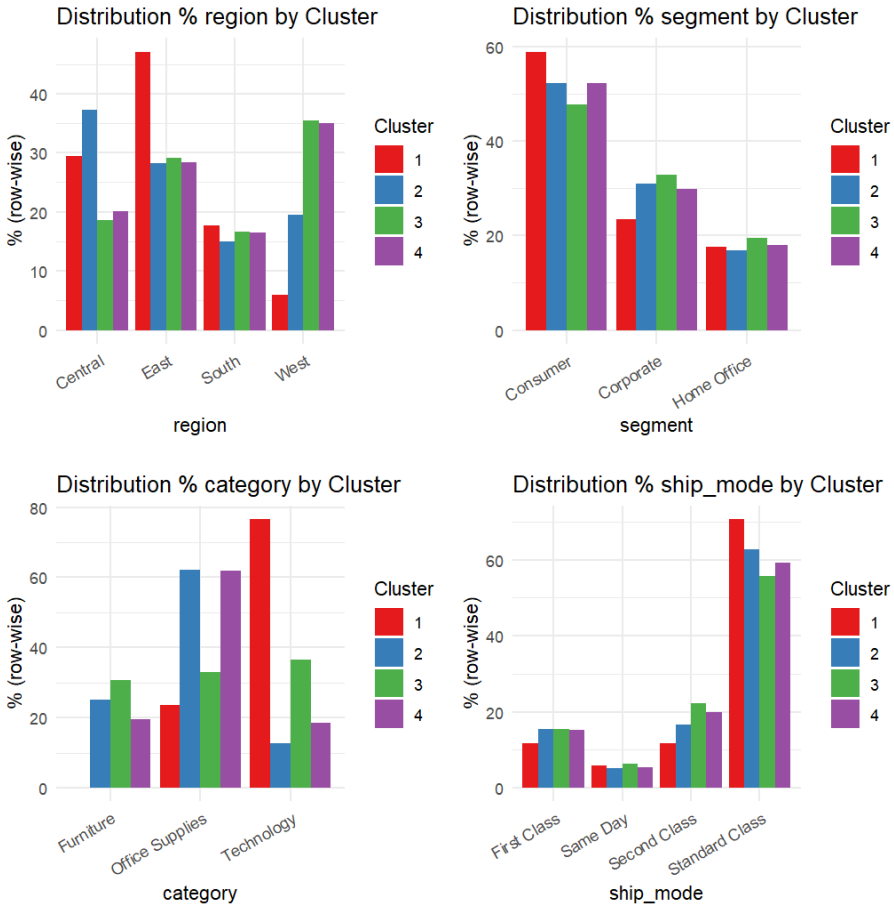

Figure 3.6: Percentage distribution of categorical variables by cluster.

- **Cluster 1: "Large Deals / Value Outliers"**
  - **Profile:** an elite group of only 17 transactions with exceptional sales volumes (mean ≈ $9,416).
  - **Outcome:** generates very high profits (mean ≈ $3,269), driven 76.5% by the *Technology* category (*Copiers* and *Machines*). Represents "big deals" with massive impact on overall results.
- **Cluster 2: "Critical Segment" (High Discounts, Losses)**
  - **Profile:** characterized by the most aggressive discount policy in the dataset (mean 45.6%).
  - **Outcome:** the only cluster with negative average profit ((-$62.17)). The *Central* region is heavily overrepresented (37.2%), indicating an area where promotional strategy destroys value.
- **Cluster 3: "High Performers" (The stars)**
  - **Profile:** transactions with high sales (mean ≈ $1,477) and very low discounts (≈ 6.8%).
  - **Outcome:** highly profitable segment (average profit > $326), strongly driven by *Technology* and *Furniture*, showing that value can be maintained without margin erosion.
- **Cluster 4: "Standard Operations" (The core business)**
  - **Profile:** the largest cluster (7,553 transactions), forming a stable revenue base.
  - **Outcome:** moderate sales (≈ $127) with low discounts (≈ 8.8%) and consistent profit (≈ $22.6), dominated (61.9%) by *Office Supplies*.

Strategic Insights

- **Discount optimization:** Cluster 2 clearly highlights where the company is losing money. Revising discount flexibility in the Central region and for loss-making categories (e.g., Binders at 38.4%) is the top priority.
- **Value focus:** Clusters 1 and 3 show that profitability depends not only on volume but on the ability to sell technology products while keeping discounts below 10%.
- **Logistics:** the distribution of `ship_mode` is similar across clusters, indicating that shipping method is not a primary driver of profitability, which instead depends almost entirely on pricing (discount) management.

This segmentation provides management with a clear map: protect the "core" (Cluster 4), promote "performers" (Cluster 3), monitor large deals (Cluster 1), and intervene decisively on the promotional policies of the "critical segment" (Cluster 2).

3.2.2 Validation and Robustness: Hierarchical Clustering

At this stage, **Hierarchical Clustering** (Agglomerative) is introduced as a validation and methodological comparison tool.
Unlike K-means, which requires specifying the number of clusters *a priori*, hierarchical clustering builds a tree structure (dendrogram) that visualizes similarity levels among transactions at different scales.

The analysis was conducted using Euclidean distance applied to principal components, providing a natural and interpretable measure of dissimilarity across continuous, scaled dimensions.

Methodological Comparison

Cluster structures were validated by comparing dendrogram partitions with K-means results.
Agreement was measured using the Adjusted Rand Index (ARI), while internal coherence was evaluated using the Silhouette Score.

Model Agreement

- **Ward Method (Ward.D2):** It is the only approach to show partial convergence with K-means (ARI ≈ 0.23). By minimizing the increase in within-cluster variance, Ward tends to create groups of comparable size. The dendrogram structure (Figure 3.7) supports a partition into 4 clusters, highlighting clear separations between divergent operational profiles.

- **Complete Linkage:** Based on the maximum distance between points, it aims to form compact clusters. However, when applied to the Superstore dataset, it showed almost no agreement with K-means (ARI < 0.05), failing to capture the overall profitability structure.

- **Average Linkage:** Although theoretically robust as it is based on average distance, in this specific use case it proves to be "misleading", failing to effectively discriminate between business segments.

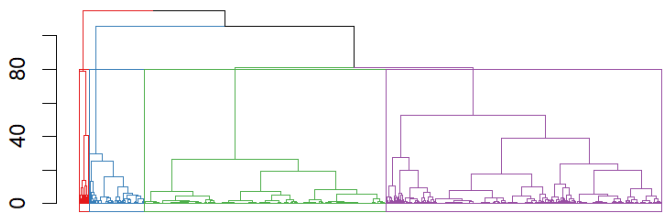

Figure 3.7: Ward.D2 Dendrogram

The Silhouette Paradox

The analysis of the Silhouette Score introduces an apparent methodological paradox: the Average and Complete methods exhibit extremely high scores (≈ 0.91 and ≈ 0.81), clearly outperforming K-means (≈ 0.54). However, a qualitative analysis of the groupings reveals that this mathematical "perfection" does not translate into operational usability, for two fundamental reasons:

- **The Outlier Trap:** Scores close to 1 arise from isolating individual extreme records into "perfect" micro-clusters in terms of distance, leaving 99% of transactions in a single indistinct macro-cluster.
- **Strategic Relevance:** A model that isolates single anomalies is useless for marketing or logistics. A manager cannot define a strategy for a cluster composed of a single transaction.

Despite a lower score, K-means (≈ 0.54) and the Ward method (≈ 0.38) generate populated and actionable segments. These groups reflect real and recurring configurations, enabling the implementation of differentiated commercial policies at scale.

Segmentation with Ward.D2

The analysis continues with the profiling of the clusters obtained using the Ward.D2 method. Having previously noted the ineffectiveness of the Complete and Average Linkage criteria—which tended to isolate individual outliers rather than providing an operational segmentation—the objective is now to verify the consistency between the Ward results and those of the K-means algorithm.
This comparison is essential to determine whether the identified segments are structural or merely dependent on the chosen algorithm. Below is the analysis of the four clusters generated by the hierarchical method:

- **Cluster 4: "Large Deals" (K-means Cluster 1)**
  - **Profile:** High-value segment (average sales $3,273) and high volumes (median 6.7 items).
  - **Product Detail:** Dominated by the *Technology* category (48%), with a massive presence of *Copiers and Machines* (12.7% and 14.5%).
- **Cluster 3: "Critical Segment" (K-means Cluster 2)**
  - **Profile:** The "financial distress" cluster, characterized by extremely high average discounts (69.5%).
  - **Operational Detail:** Records a heavily negative average profit (-$114.65). It is composed of 72% *Office Supplies*, but the most significant data emerges at the sub-sector level: 65% of transactions involve *Binders*.
  - **Comparison:** Ward identifies the "black hole" of profitability with surgical precision. Compared to K-means, the link between extreme discounts on *Binders* and economic loss appears even clearer and more localized here.
- **Cluster 2: "High-Volume Retail" (K-means Cluster 3)**

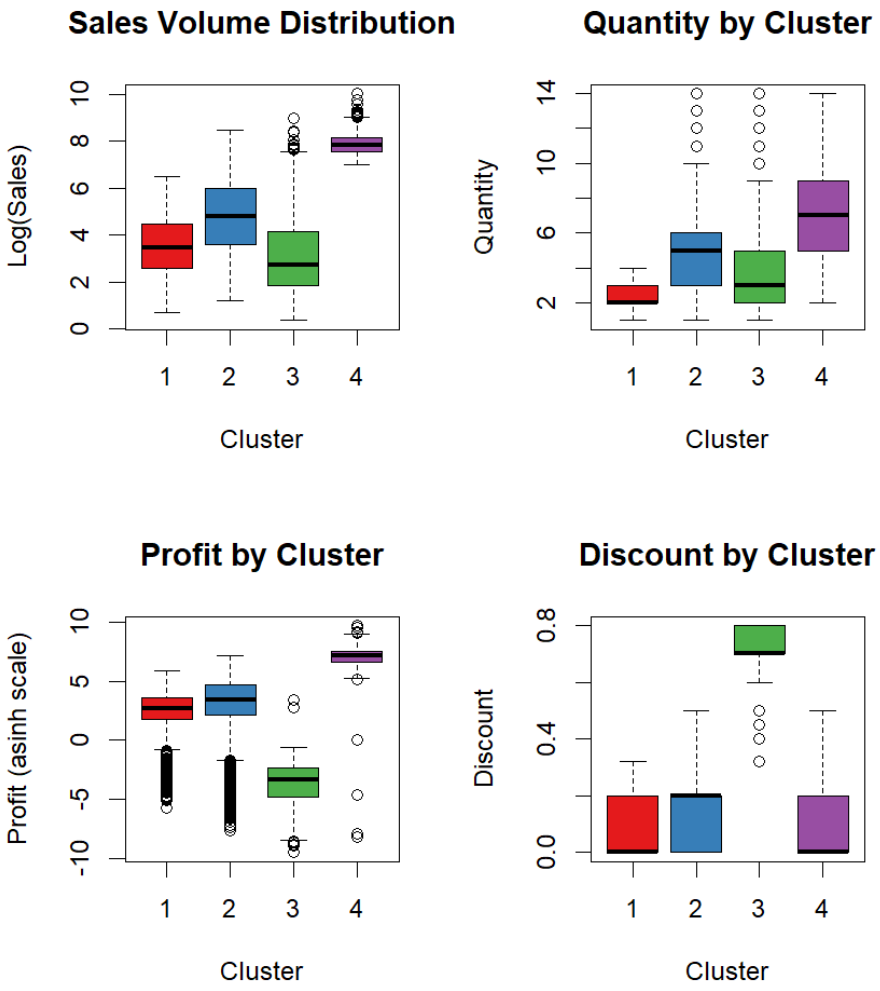

Figure 3.8: Cluster characterization and contribution of numerical variables.

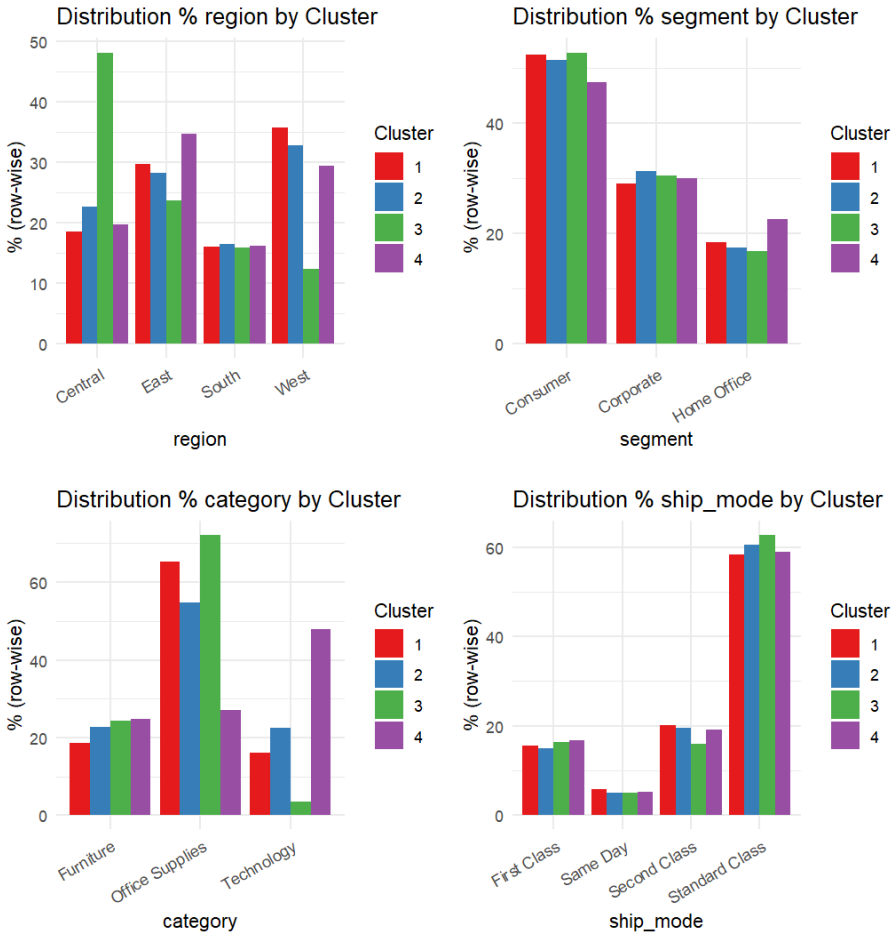

Figure 3.9: Percentage distribution of categorical variables by cluster.

- **Profile:** Transactions with solid average sales ($279) and a good quantity of items per order (average 4.9).
- **Strategic Detail:** *Discounts* are moderate (13.6%) and profitability is positive ($38). A balanced mix of *Furniture, Office Supplies, and Technology* is observed.
- **Comparison:** This represents the medium-high market range. While K-means tended to isolate brighter "stars," Ward groups here the *Corporate and Consumer* segments that consistently purchase furniture and telephony.

- **Cluster 1: "Core Transactions" (K-means Cluster 4)**
  - **Profile:** The company's daily "engine" (4,153 transactions), focused on small orders (average $66).
  - **Operational Detail:** Minimal discounts (5.9%) and modest but steady profit ($14). Dominated by *Paper, Art, and Furnishings.*
  - **Comparison:** This is the equivalent of the "Heart of the Business" found in K-means. The consistency between the two methods for this group (the largest one) validates the stability of the company's transactional base.

Robustness Conclusions

The systematic comparison between the results obtained through the Ward.D2 method and those derived from K-means reveals significant analytical stability: both algorithms, despite operating with different mathematical logics—hierarchical and partitional, respectively—identify the same latent structures within the *Superstore* dataset. This methodological convergence allows for definitive conclusions regarding the robustness of the analysis: the significant overlap between the two models demonstrates that the identified segments are not algorithmic or random artifacts, but rather represent real and recurring configurations of sales data. The stability of these groupings (particularly for the critical loss-making and high-value clusters) provides a solid and reliable empirical basis for the implementation of future operational strategies and managerial decisions.

3.3 Conclusions and Key Results

The analysis conducted in the third chapter confirms that the Superstore's profitability is not a random phenomenon but responds to well-defined latent structures. The main results can be summarized as follows:

- **Effectiveness of Dimensionality Reduction:** The PCA analysis demonstrated that transaction complexity can be drastically reduced: with just two principal components (*PC1* "Performance" and *PC2* "Volume Strategy"), 66% of the total variance is explained, clearly isolating the conflict between increasing volumes (driven by discounts) and maintaining profit margins.

- **Identification of the Critical Segment:** Clustering allowed for the precise mapping of the so-called "black hole" of profitability (*Cluster 2*). This segment

is characterized by extremely high average discounts (over 45–69%) that systematically lead to significant average losses, with a particular impact in the *Central* region and the *Binders* sub-category.

- **Strategic Value of Top Performers:** At the opposite extreme, the high-value clusters (*Cluster 1* and *Cluster 3*) demonstrate that it is possible to generate excellent profits by focusing on the *Technology* category and keeping discount thresholds below 10%, regardless of absolute volumes.

For management, the absolute priority lies in revising pricing policies for the critical segment, as the analysis shows that high volumes in this area are not synonymous with success, but rather with value destruction.

Chapter 4

Supervised Analysis

Having concluded the exploration of the intrinsic structure of the data, the analysis transitions toward supervised learning (Supervised Learning). In this phase, the objective shifts from describing the dataset to building predictive models capable of anticipating specific events, with a primary focus on preventing economic losses.

4.1 Introduction and Configuration

The core of the analysis is the binary target variable `loss_flag`, created to identify whether a transaction has generated a negative profit. Identifying these transactions in advance allows management to intervene proactively on potentially harmful sales.

To ensure statistical robustness, the following pre-processing procedures have been implemented:

- **Feature Engineering:** the variable `sales` has been subjected to a log-transformation (`log_sales`) to normalize the distribution and mitigate the impact of outliers;
- **Variable Selection:** excessively granular variables such as `city` or `state` and `sub_category` have been removed to prevent overfitting.
- **Stratified Partitioning:** the dataset has been divided into a training set (70%) and a test set (30%). Stratification ensures that the proportion of loss-making transactions is identical in both sets, avoiding bias in evaluations.
- **Baseline Management (Reference Levels):** to make the coefficients of the models (such as logistic regression) interpretable, the most frequent category for each categorical variable has been set as the reference level (e.g. *West* for the region, *Standard Class* for shipping). This allows measuring the impact of other categories relative to the company's operational standard.

4.1.1 Integration of Unsupervised Segmentation

Although the unsupervised analysis identified clear and informative business segments, including clusters in supervised models for loss prediction (`loss_flag`) worsens both the overall quality and the parsimony of the model.

- **AIC Deterioration (Akaike Information Criterion):** The AIC rewards models that better explain the data with the fewest variables.
  - Base model: 2179.6
  - Model with clusters: 2260.4

  An increase of 81 points is statistically significant and indicates that clusters introduce noise or unnecessary complexity without improving the model fit to the data.

- **Degradation of variable significance:** In the model with clusters, some previously significant variables or trends related to `segment` lose statistical relevance (increase in *p-value*). This occurs because clusters are built from `sales`, `discount` and `quantity`. By including both the original variables and the clusters, *multicollinearity* is generated: the same information is provided twice, making the estimates less interpretable.

- **Coefficient instability (Standard Error):** The standard error of the intercept increases from 0.32 to 1.78, a typical signal of model instability: clusters absorb variance that would otherwise be attributed to other covariates, worsening the reliability of the estimates.

Although they are not suitable as explanatory variables in the final logistic model, clusters remain useful for business strategy: they provide an operational and communicable interpretation, namely a name and a profile to risk groups (qualitative analysis). For example, identifying a "critical cluster" facilitates the identification of transactions to act upon without having to directly interpret the logistic equation.

4.1.2 Evaluation Metrics (Model Performance)

In a classification context, accuracy alone is often a partial metric. To ensure a rigorous and multidimensional evaluation of the predictive capability of the models, the following metrics have been adopted:

- **Precision:** measures the robustness of positive predictions, indicating the proportion of transactions correctly identified as loss-making relative to the total number of signals produced by the model.

- **Recall (Sensitivity):** expresses the model's ability to systematically capture critical transactions, calculating the share of actual losses correctly identified out of the total cases present in the dataset.

- **F1-Score:** represents the harmonic mean between *precision* and *recall*, providing a fundamental synthetic indicator to balance model effectiveness when the costs of false positives and false negatives differ.

- **Accuracy:** measures the proportion of correct predictions over the total predictions. Accuracy can be misleading in imbalanced datasets, as the majority class may dominate, making accuracy appear high even when the minority class is poorly predicted.

- **AUC (Area Under the ROC Curve)** reflects the model's ability to distinguish between classes, with higher values indicating better separation. AUC is less affected by class imbalance, making it a reliable metric for evaluating model performance in imbalanced datasets.

From an operational perspective, Recall is considered the main optimization driver. In a risk mitigation framework, the primary objective is to minimize false negatives: the opportunity cost of a "false alarm" (a transaction flagged in advance that does not generate a loss) is significantly lower than the economic damage resulting from a loss-making transaction not detected by the system.

4.1.3 Algorithm Selection

The methodological approach follows an incremental path: starting from linear statistical models to ensure maximum interpretability, up to the use of advanced machine learning techniques (Machine Learning) to maximize predictive capability.

Algorithm	Rationale for use
Logistic Regression	Reference model for binary classification, adopted to ensure interpretability and transparency.
Regularization (LASSO / Ridge)	Extension of logistic regression with penalty terms: LASSO promotes automatic variable selection; Ridge reduces variance in the presence of multicollinearity, helping to contain overfitting.
Random Forest	Ensemble of trees trained on bootstrap samples and random subsets of variables: used as a robust and stable benchmark, capable of capturing non-linearities and interactions.
GBM (Gradient Boosting Machine)	Boosting algorithm that combines weak trees sequentially by optimizing a loss function: used to increase predictive capability on complex patterns and non-linear relationships, with overfitting control through learning rate and tree depth.
XGBoost	Optimized implementation of gradient boosting with regularization and efficient handling of complex datasets: adopted to maximize predictive performance and generalization, maintaining flexibility in modeling non-linearities and interactions.

4.2 Results Analysis

Knowing that the objective is the prediction of `loss_flag` (i.e., identifying whether a *Superstore* transaction will end in loss or profit) drastically changes the strategic priorities of the analysis: in a retail and supply chain context, a false alarm (saying

that a sale is at a loss when it is not) may block a valid commercial operation, but failing to identify a transaction that generates a strong loss (false negative) is a direct cost that erodes company margins.

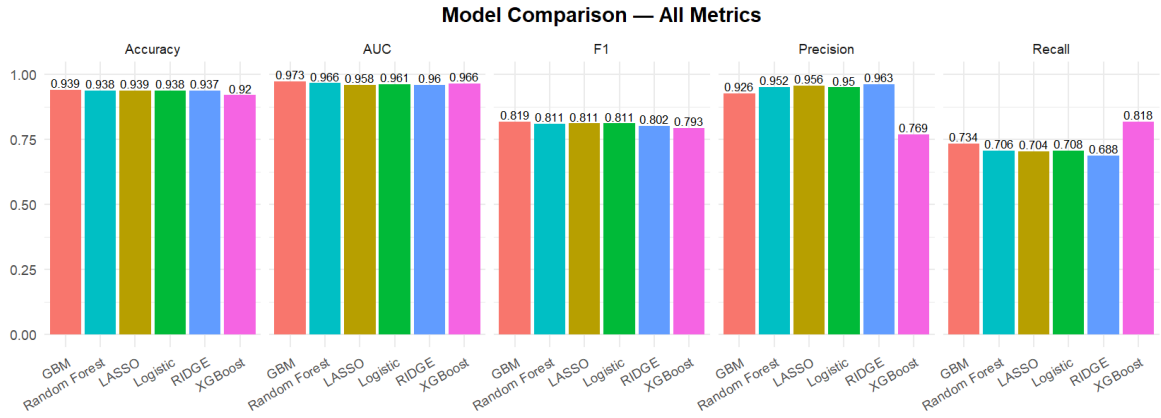

Figure 4.1: Comparison metrics

The application of the models to the Superstore dataset reveals an interesting dynamic: the signal related to losses is extremely clean, as demonstrated by an average Accuracy of 94% and an AUC approaching perfection with 0.973 for the GBM. In this scenario, key variables — such as the applied discount (`Discount`), the product category, and shipping costs (Figure 4.2) — exert a decisive influence. The discount (Figure 4.2), in particular, is often the main "trigger" of loss: beyond a certain threshold, the transaction mathematically turns negative, explaining why linear models (LASSO, Logistic) achieve results almost identical to complex tree-based models.

The Strategic Winner: XGBoost

Although in purely statistical analysis GBM appears superior in terms of balance, in a business context oriented toward margin protection, the actual winner becomes XGBoost.

- XGBoost shows the highest Recall (0.818) in the set. In business terms, this means that XGBoost is the model capable of intercepting 82% of loss-making transactions before they occur.
- It sacrifices Precision (0.769), which implies a share of 'false positives" (transactions flagged as risky that would actually have been profitable). However, for a Superstore manager, it is often preferable to manually review 20% of 'suspicious" orders rather than miss outright losses that more 'conservative" models such as RIDGE (Recall 0.688) would not detect.

Conservative Efficiency: LASSO and RIDGE

LASSO and RIDGE regularization models confirm themselves as tools of surgical precision, with Precision values between 0.95 and 0.96. These models are ideal if the company intends to implement an automatic transaction blocking system without human intervention: when LASSO says 'loss", it is correct in almost 96% of cases.

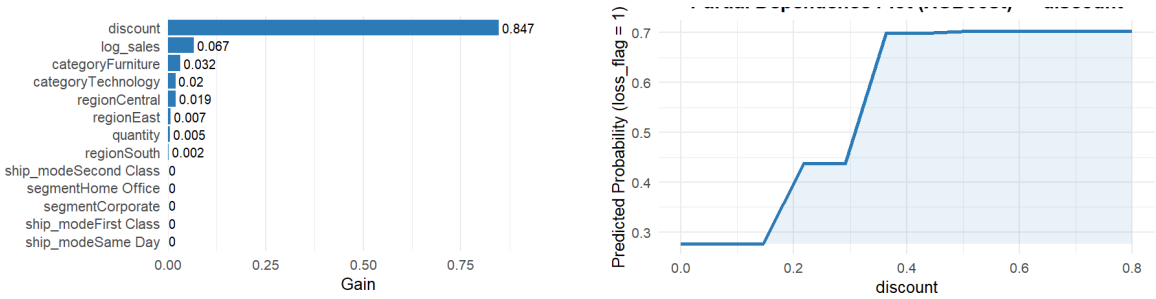

Figure 4.2: XGBoost analysis. *Left*: Variables importance. *Right*: Partial dependence Plot (XGBoost) for discount.

The limitation remains Recall (≈ 0.70): they ignore about 30% of loss-making transactions, leaving an exposed flank that could weigh on the quarterly balance.

The Worst: RIDGE

Despite high precision, the RIDGE model falls to the last place in the functional ranking with a Recall of 0.688, the lowest recorded. In a Loss Detection problem, a model that fails to detect nearly one-third of total losses is considered inefficient, since the primary objective is not certainty of prediction, but maximum coverage of financial risk.

4.3 Conclusions and Key Results

The integrated analysis has made it possible to isolate the determining factors for the economic health of the Superstore, leading to the following key results:

- **The Dominant Driver:** The discount (`Discount`) ‘e the critical variable, capable of explaining on its own 84% of the variance in profitability; beyond the 20–30% threshold, profit systematically declines (Figure 4.2).
- **Risk Areas:** The *Furniture* category and the *Central* region are structurally less profitable and require immediate intervention in pricing policies.
- **Predictive Effectiveness:** The XGBoost model demonstrated the highest capability in detecting loss-making transactions (Recall $\approx$ 0.82), while LASSO regularization confirmed that a parsimonious structure of only four variables ‘e sufficient for accurate risk management.

In conclusion, moving from a superficial reading to an analytical approach allows transforming invisible inefficiencies into opportunities for savings and marginal growth.

4.4 Possible Implementations and Future Developments

To transform this study into a dynamic decision-making tool, the following evolutions based on the enrichment of the information assets are proposed:

- **Analysis of 'Loss Leader Products" (*Loss Leader Strategy*):** The integration of historical customer data would allow verifying whether loss-making transactions (such as those in the 'Critical Segment") are actually marketing moves. By cross-referencing orders, it would be possible to determine whether an aggressive discount on a product (e.g. *Tables*) has generated over time high-margin purchases in other categories (e.g. *Technology*), justifying the initial loss as a customer acquisition cost.

- **Customer Lifetime Value (CLV) Modeling:** Instead of analyzing the profitability of the single transaction, implementing CLV models would allow weighting `loss_flag` based on customer loyalty. A loss on a long-standing "Corporate" customer has a different strategic value compared to a loss on an occasional ("Consumer") customer.

- **Increase in Analytical Granularity:** In the present study, some dimensions were aggregated to ensure a clear and synthetic overview. In the future, the analysis may benefit from a deeper *drill-down* approach, going into the detail of variables such as `City`, `State` and `Sub-Category`. This would allow digging into the core of the business, isolating specific micro-markets or product niches that are currently averaged within regional or categorical aggregates, thus enabling the extraction of highly precise and localized prescriptive *insights*.

- **Demand Elasticity Analysis:** Using discount data to calculate price elasticity for each sub-category. This would make it possible to identify where discounts are truly necessary to drive volume and where, instead, price reductions do not generate sufficient sales increase to offset margin loss.

- **Geofencing and Predictive Logistics:** Given that the *Central* region shows negative anomalies independent of the product, it would be useful to integrate data on actual shipping costs and local competition. The objective è to understand whether the loss is due to a territorial "price war" or to supply chain inefficiencies in that specific area.

Chapter 5

Bibliography

https://www.kaggle.com/datasets/ibrahimelsayed182/superstore/data